SENIOR HONORS THESIS · MAY 2026

Market Concentration as a Signal for *Tail Risk*

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REASEARCH QUESTION | Does rising concentration in high-Domar-weight sectors lead the *repricing of tail risk* in equity options markets?

Background

> *A handful of factories in Taiwan produce nearly all of the world's most advanced chips. A single Dutch firm builds every machine capable of etching them. The global economy has knowingly assembled this kind of dependency, and it should be visible in financial market prices.*

Recent history supports the claim. The **Tōhoku earthquake** (2011) shut down a single Renesas plant and cascaded into halted auto lines across North America. The **Ever Given** grounding (2021) rippled through container shipping for months. Concentrated networks are highly efficient most of the time and catastrophically fragile when a hub fails.

The same logic now characterizes the U.S. equity market itself. The seven largest firms account for roughly one-third of the S&P 500. Through the wealth effect, a sharp reversal in these dominant firms would translate quickly into a broad pullback in household spending across the economy.

Whether the current configuration represents peak efficiency or a systemic vulnerability turns on whether options markets correctly price the crash risk that concentrated production networks create. The **25-delta risk reversal,** the difference between out-of-the-money put and call implied volatilities, isolates exactly this asymmetric tail-pricing moment.

Theory

Production networks generate endogenous negative skewness whenever inputs are **complements** ($\sigma < 1$). A second-order Taylor expansion of aggregate output around the steady state breaks the linearity of Hulten's theorem:

EQ. 1 · OUTPUT EXPANSION

$$\Delta y \approx \Sigma_i \lambda_i \varepsilon_i + \tfrac{1}{2} \Sigma_{i,j} \left(\partial^2 y / \partial\varepsilon_i \partial\varepsilon_j\right) \varepsilon_i \varepsilon_j$$

First term: Hulten's linear, symmetric baseline. Second term: non-linear network amplification.

EQ. 2 · DIAGONAL CURVATURE

$$\partial^2 y / \partial\varepsilon_i^2 = \lambda_i \,(1 - \lambda_i)\, (\sigma - 1)$$

When $\sigma < 1$, the term is strictly negative. High concentration on hub sectors (large λ_i, equivalently large HHI) multiplied by strong complementarity produces large negative second-order terms — **mechanically manufacturing negative skewness from symmetric shocks.**

EQ. 3 · NON-MONOTONIC FRAGILITY

$$\partial / \partial HHI \; \mathbb{E}[\Delta y \mid \|\varepsilon\|] \begin{cases} \geq 0 \text{ if } \|\varepsilon\| < \bar{\varepsilon} \text{ stabilizing branch} \\ < 0 \text{ if } \|\varepsilon\| > \bar{\varepsilon} \text{ amplifying branch} \end{cases}$$

Hub formation absorbs ordinary fluctuations efficiently but channels extreme shocks. The boundary is governed by σ — the substitutability of inputs.

Data & Empirics

A monthly panel of the **eleven Select Sector SPDR ETFs** from Jan 2005 through Feb 2026 combines firm-level constituent data, option-implied volatility surfaces, and BEA Input-Output tables.

11	247	2,425	4
GICS Sectors	Months of observations	Sector-month observations	Sources: Bloomberg, BEA, OptionMetrics, FRED

Outcome: The one-month-ahead 25-delta risk reversal, $RR_{j,t+1}$ = IV(OTM put) − IV(OTM call). A more negative value signals investors are paying a larger premium for crash protection than for upside exposure.

Supplier HHI: For each downstream sector, the I-O-weighted concentration of every upstream sector it depends on. Captures inherited supply-chain fragility.

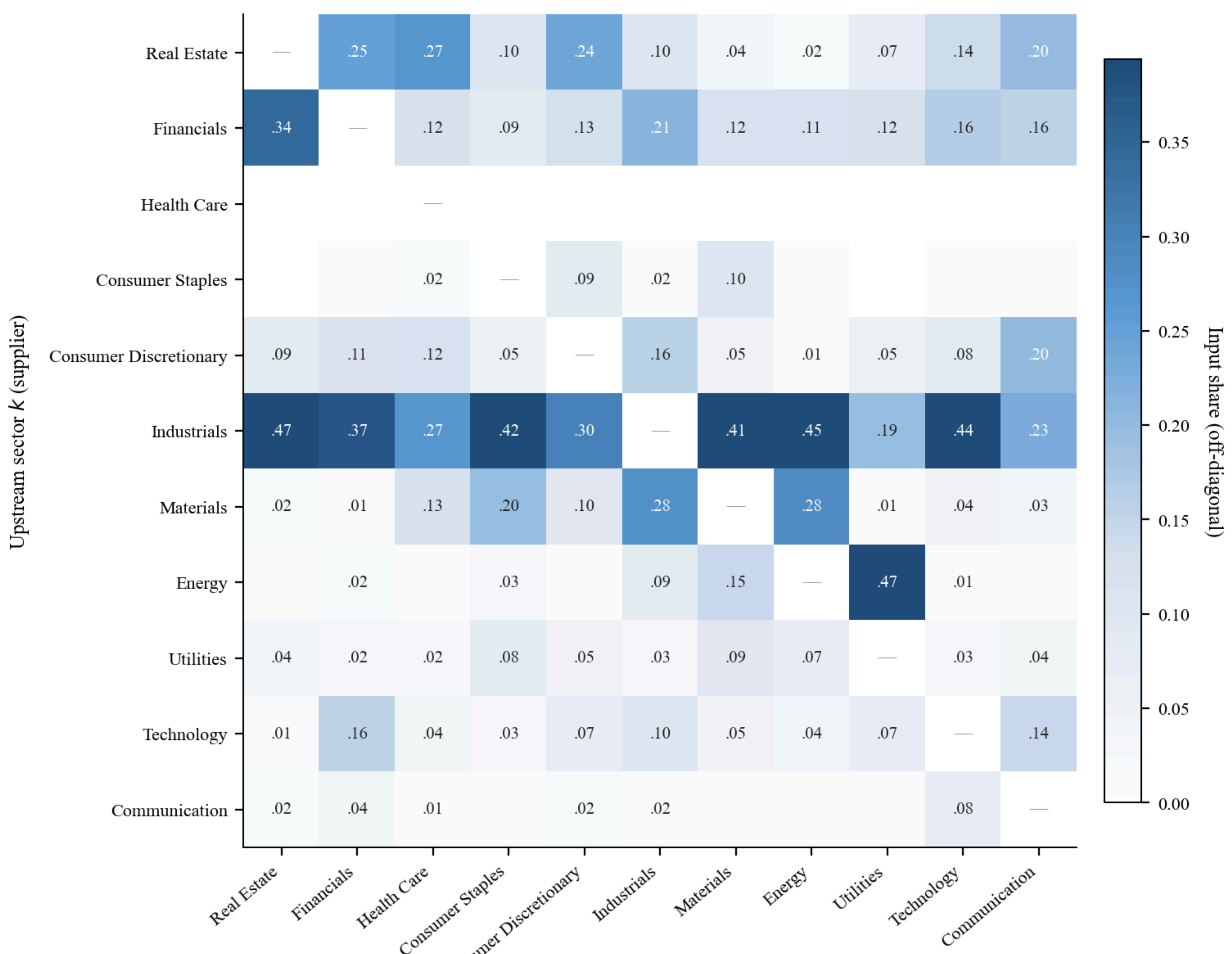

Figure 1. Inter-sector input-output weight matrix $\omega_{j,k}$ from the 2024 BEA tables. Each cell gives the share of downstream sector j's (column) inputs sourced from upstream sector k (row). **Industrials** dominates as a universal upstream supplier; the **Energy → Utilities** dependency (.47) is the largest single bilateral link.

MODEL 3A · HETEROGENEOUS HHI LOADING

$$RR_{j,t+1} = \alpha_j + \gamma_t + \Sigma_j \beta_{1,j} \left(HHI_{j,t} \times D_j\right) + \beta_2 \, SupplierHHI_{j,t} + \theta' X_{j,t} + \varepsilon_{j,t+1}$$

Sector dummies D_j let the loading on internal HHI vary, recovering the substitutability gradient predicted by Eq. 2.
Robustness: Driscoll-Kraay SEs (bw = 6, 12); leave-one-year-out cross-validation across 15 calendar-year folds.

Results

The Input-Output-weighted Supplier HHI is the dominant homogeneous predictor of the one-month-ahead risk reversal. Internal HHI is insignificant when homogeneous, but a sector-heterogeneous specification reveals a sharp economic gradient that the homogeneous estimate averages away.

BASELINE STRUCTURAL MODEL

−0.076 (t = −5.22)

Supplier HHI coefficient on the forward 25-delta risk reversal, significant at the 1% level under both entity-clustered and Driscoll-Kraay standard errors. Consistent with the stabilizing branch of the nonmonotonic mechanism.

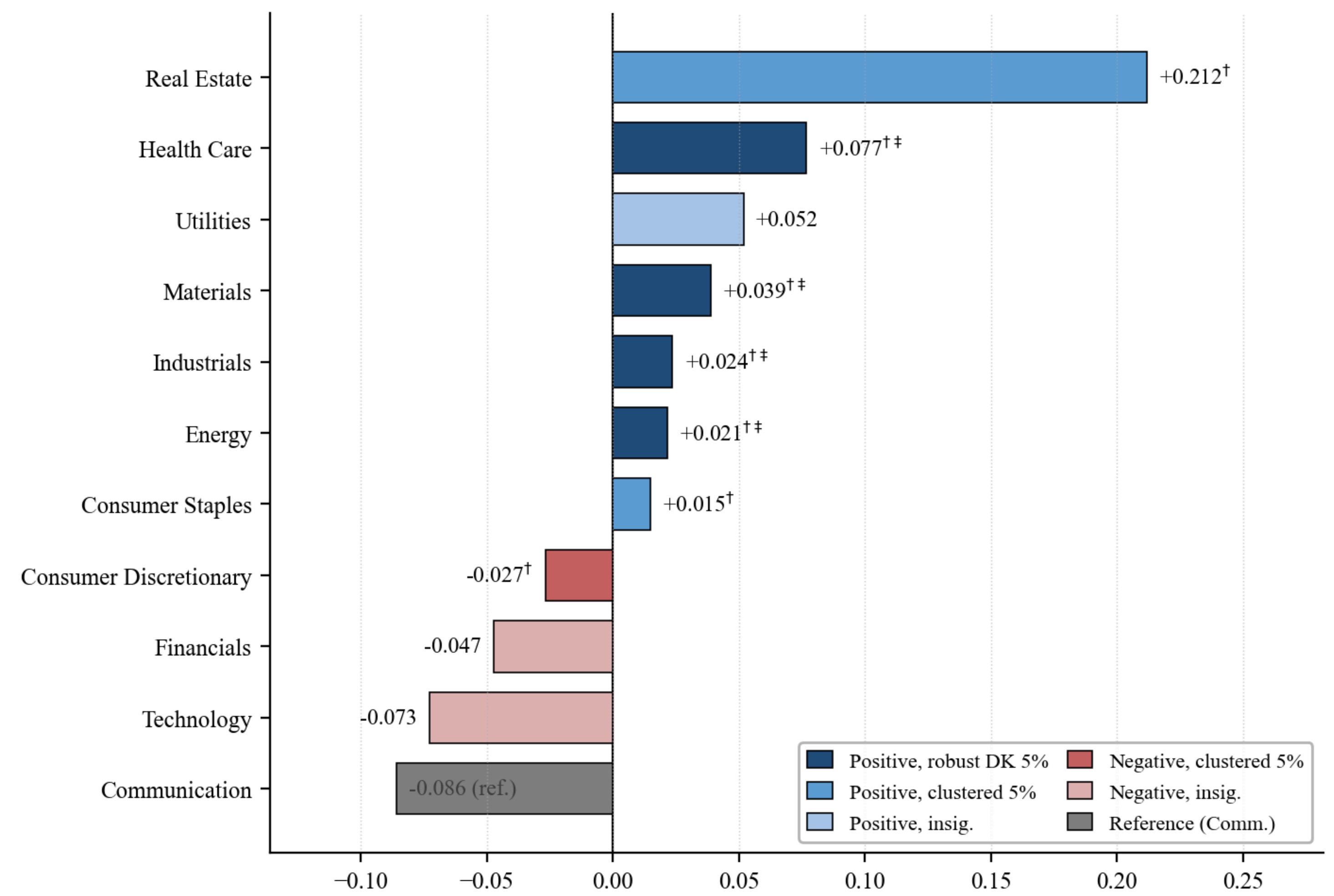

Figure 2. Net sector-specific HHI effects under Model 3A (heterogeneous internal concentration). **Seven physical-capital-intensive sectors** load positively on HHI; **digital sectors** load negatively. † marks 5% entity-clustered significance; ‡ marks robustness to Driscoll-Kraay (bw = 6).

> *The **substitutability gradient** is the headline empirical finding. It maps directly to the Baqaee & Farhi (2019) prediction that complementarities, not concentration alone, drive non-linear amplification.*

AMPLIFYING · $\sigma < 1$

Physical-capital-intensive sectors

Real Estate	+9.9 bp
Energy	+9.8 bp
Health Care	+8.5 bp
Industrials	+7.5 bp
Materials	+6.3 bp
Consumer Staples	+3.3 bp
Utilities	+1.8 bp

MUTED · $\sigma \approx 1$

Digital & platform sectors

Communication	−23.7 bp
Cons. Discretionary	−13.4 bp
Technology	−12.4 bp
Financials	−2.1 bp

$\Delta RR_{j,t+1}$ per +1σ HHI move; basis points.

FUTURE WORK Direct empirical estimation of sector-level σ combined with BEA gross-output Domar weights · firm-level customer-supplier disclosures to resolve network topology below sector aggregation · threshold-regression diagnostics to locate the boundary $\bar{\varepsilon}$ between stabilizing and amplifying branches · realized-skewness validation of the option-implied dependent variable.